

Branching

Boolean expressions can be used to determine which route a program should follow, depending on whether the answer to the expression is “True” or “False”. This is known as “branching”.

SEE ALSO

◀ 64–65 Decisions and branches

◀ 118–119 Making decisions

Do or do not

The “if” command means that if a condition is “True”, then the program runs a block of commands. If the condition isn’t “True”, the block is skipped. The block after the “if” command is always indented by four spaces.

1 “if” condition

This code asks the user if it’s their birthday. It checks whether the answer is “y”. If so, a birthday message is printed.

Indented by four spaces

```
ans = input('Is it your birthday? (y/n)')
if ans == 'y':
    print('Happy Birthday!')
```

Prompts users what to type in

This part of the program only runs if the user types “y”

2 Output if condition is “True”

Run the program and enter “y”. The message is printed. It doesn’t appear if anything else is entered.

Type in “y”

```
Is it your birthday? (y/n)y
Happy Birthday!
```

The message appears

Do this or that

The “if” command can be combined with an “else” command. This combination means that if something is “True”, one thing happens, and if not, something else happens.



1 “if-else” condition

If “y” is entered, the program prints a special message for New Year. It shows a different message if anything else is entered.

Remember to put a colon here too

```
ans = input('Is it New Year? (y/n)')
if ans == 'y':
    print('Happy New Year!')
    print('Time for Fireworks.')
else:
    print('Not yet!')
```

Remember the colon

This message only appears if the user enters “y”

Only runs if user does not enter “y”